Statistical Foundations of ML — Formula Sheet

1 Foundations

Definition 1.1. Sample space S: set of all outcomes. An event is any $A \subseteq S$.

Definition 1.2. A probability measure P satisfies

- $P(S) = 1$,
- $P(A) \geq 0$,
- $P(\bigcup_i A_i) = \sum_i P(A_i)$ for pairwise disjoint A_i.

$$\boxed{P(A^c) = 1 - P(A) \qquad P(\varnothing) = 0 \qquad 0 \leq P(A) \leq 1}$$

$$\boxed{A \subseteq B \implies P(A) \leq P(B)}$$

$$\boxed{P(A \setminus B) = P(A) - P(A \cap B)}$$

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

Inclusion–Exclusion (3 events)

$$\boxed{\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C) \end{aligned}}$$

Union bound

$$\boxed{P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)}$$

Bonferroni inequalities

Let $S_1 = \sum_i P(A_i)$, $S_2 = \sum_{i<j} P(A_i \cap A_j)$, and $S_3 = \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$. Then

$$\boxed{S_1 - S_2 \leq P\left(\bigcup_i A_i\right) \leq S_1, \qquad P\left(\bigcup_i A_i\right) \leq S_1 - S_2 + S_3.}$$

Stopping inclusion–exclusion after an odd term gives an upper bound; after an even term, a lower bound.

2 Conditional Probability & Independence

Definition 2.1.

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0}$$

Multiplication rule

$$\boxed{P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)}$$

Chain rule

$$\boxed{\begin{aligned} P(A_1 \cap \cdots \cap A_n) = {} & P(A_1)P(A_2 \mid A_1) \cdots \\ & P(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \end{aligned}}$$

Law of total probability

If $B_1, \ldots, B_n$ form a partition of S,

$$\boxed{P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)}$$

Bayes' rule

$$\boxed{P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}}$$

Definition 2.2. A and B are independent iff

$$\boxed{P(A \cap B) = P(A)P(B) \iff P(A \mid B) = P(A)}$$

The events $A_1, \ldots, A_n$ are mutually independent if the product rule holds for every nonempty subset of the events.

3 Counting & Sampling

$$\boxed{n! = 1 \cdot 2 \cdots n \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}}$$

Choosing k items from n:

	Without replacement	With replacement
Ordered	$\dfrac{n!}{(n-k)!}$	n^k
Unordered	$\binom{n}{k}$	$\binom{n+k-1}{k}$

- Arrangements of n objects: $n!$; circular arrangements: $(n-1)!$.
- Binomial theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.
- Equally likely outcomes: $P(A) = |A|/|S|$.

$$\boxed{\binom{n}{n_1, \ldots, n_r} = \frac{n!}{n_1! \cdots n_r!}, \qquad \sum_{i=1}^{r} n_i = n}$$

4 Random Variables & CDF

Definition 4.1. A random variable X is a real-valued function on the sample space.

Definition 4.2. $F_X(x) = P(X \leq x)$.

$$\boxed{P(a < X \leq b) = F_X(b) - F_X(a) \qquad P(X > a) = 1 - F_X(a)}$$

F_X is nondecreasing and right-continuous, with $F_X(-\infty) = 0$ and $F_X(\infty) = 1$.

$$\boxed{P(X = x) = F_X(x) - F_X(x^-), \qquad F_X(x^-) = \lim_{t \uparrow x} F_X(t)}$$

5 Discrete Random Variables & PMF

Definition 5.1. $p_X(x) = P(X = x)$.

$$\boxed{p_X(x) \geq 0 \qquad \sum_x p_X(x) = 1 \qquad F_X(t) = \sum_{x \leq t} p_X(x)}$$

$$\boxed{P(X \in A) = \sum_{x \in A} p_X(x)}$$

	$p_X(x)$	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	$p^x(1-p)^{1-x},\ x \in \{0,1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$
Geometric(p)	$p(1-p)^{x-1},\ x \geq 1$	$1/p$	$(1-p)/p^2$
Poisson(λ)	$e^{-\lambda}\lambda^x/x!,\ x \geq 0$	λ	λ

- If $X \sim \mathrm{Geo}(p)$, then $P(X > k) = (1-p)^k$.
- Memorylessness: $P(X > m+n \mid X > m) = P(X > n)$.
- A binomial random variable counts successes in n independent Bernoulli(p) trials.

6 Continuous Random Variables & PDF

Definition 6.1. A PDF satisfies

$$\boxed{f_X(x) \geq 0 \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1}$$

$$\boxed{P(a \leq X \leq b) = \int_a^b f_X(x)\,dx}$$

$$\boxed{F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt \qquad f_X(x) = F_X'(x) \qquad P(X = x) = 0}$$

	$f_X(x)$	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Uniform(a, b)	$1/(b-a),\ a \leq x \leq b$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$\lambda e^{-\lambda x},\ x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

- Uniform: $F_X(x) = (x-a)/(b-a)$ for $a \leq x \leq b$.
- Exponential: $F_X(x) = 1 - e^{-\lambda x}$, $P(X > t) = e^{-\lambda t}$.
- Exponential memorylessness: $P(X > s+t \mid X > s) = P(X > t)$.
- If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = (X-\mu)/\sigma \sim \mathcal{N}(0,1)$, $F_X(x) = \Phi((x-\mu)/\sigma)$, and $\Phi(-z) = 1 - \Phi(z)$.

7 Expectation & Variance

$$\boxed{\mathbb{E}[g(X)] = \sum_x g(x)p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int g(x)f_X(x)\,dx}$$

$$\boxed{\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c}$$

$$\boxed{\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}[X_i] \qquad \mathbb{E}[I_A] = P(A)}$$

If X and Y are independent,

$$\boxed{\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]}$$

Variance

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}$$

$$\boxed{\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)}$$

If X_i are independent,

$$\boxed{\mathrm{Var}\left(\sum_i X_i\right) = \sum_i \mathrm{Var}(X_i)}$$

8 Random Vectors, Joint CDF & Independence

Definition 8.1.

$$\boxed{F_{XY}(x, y) = P(X \le x, Y \le y)}$$

$$\boxed{F_X(x) = \lim_{y\to\infty} F_{XY}(x, y) \qquad F_Y(y) = \lim_{x\to\infty} F_{XY}(x, y)}$$

Rectangle probability

$$\boxed{\begin{aligned}P(a < X \le b,\ c < Y \le d) &= F_{XY}(b, d) - F_{XY}(a, d)\\ &\quad - F_{XY}(b, c) + F_{XY}(a, c)\end{aligned}}$$

Definition 8.2. X and Y are independent iff

$$\boxed{F_{XY}(x, y) = F_X(x)F_Y(y) \quad \text{for all } x, y}$$

9 Discrete Random Vectors

Definition 9.1. $p_{XY}(x, y) = P(X = x, Y = y)$.

$$\boxed{p_{XY}(x, y) \ge 0 \qquad \sum_x \sum_y p_{XY}(x, y) = 1}$$

$$\boxed{p_X(x) = \sum_y p_{XY}(x, y) \qquad p_Y(y) = \sum_x p_{XY}(x, y)}$$

$$\boxed{p_{X|Y}(x \mid y) = \frac{p_{XY}(x, y)}{p_Y(y)}, \qquad p_Y(y) > 0}$$

$$\boxed{p_X(x) = \sum_y p_{X|Y}(x \mid y)p_Y(y)}$$

$$\boxed{p_{XY}(x, y) = p_X(x)p_Y(y) \quad \text{if independent}}$$

$$\boxed{\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y)p_{XY}(x, y)}$$

10 Jointly Continuous Random Vectors

Definition 10.1. $f_{XY}(x, y) \ge 0$,

$$\boxed{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x, y)\,dx\,dy = 1}$$

$$\boxed{P((X, Y) \in A) = \iint_A f_{XY}(x, y)\,dx\,dy}$$

$$\boxed{f_{XY}(x, y) = \frac{\partial^2}{\partial x\,\partial y} F_{XY}(x, y)}$$

$$\boxed{f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\,dy \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\,dx}$$

$$\boxed{f_{X|Y}(x \mid y) = \frac{f_{XY}(x, y)}{f_Y(y)}, \qquad f_Y(y) > 0}$$

$$\boxed{f_{XY}(x, y) = f_X(x)f_Y(y) \quad \text{if independent}}$$

$$\boxed{\mathbb{E}[g(X, Y)] = \iint g(x, y)f_{XY}(x, y)\,dx\,dy}$$

11 Covariance & Correlation

$$\boxed{\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}$$

$$\boxed{\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}$$

$$\boxed{\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X) \qquad \mathrm{Cov}(X, X) = \mathrm{Var}(X)}$$

$$\boxed{\mathrm{Cov}(aX + b, cY + d) = ac\ \mathrm{Cov}(X, Y)}$$

$$\boxed{\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X, Y)}$$

$$\boxed{\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)}$$

$$\boxed{\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}, \qquad -1 \le \mathrm{Corr}(X, Y) \le 1}$$

- Independent $\implies \mathrm{Cov}(X, Y) = 0$.
- In general, $\mathrm{Cov}(X, Y) = 0 \not\implies$ independent.

12 Bivariate Gaussian

$$\boxed{\begin{pmatrix}X\\Y\end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix}\mu_X\\\mu_Y\end{pmatrix}, \begin{pmatrix}\sigma_X^2 & \rho\sigma_X\sigma_Y\\ \rho\sigma_X\sigma_Y & \sigma_Y^2\end{pmatrix}\right)}$$

$$\boxed{X \sim \mathcal{N}(\mu_X, \sigma_X^2) \qquad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)}$$

$$\boxed{\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y \qquad \mathrm{Corr}(X, Y) = \rho}$$

$$\boxed{\begin{aligned}aX + bY \sim \mathcal{N}\big(&a\mu_X + b\mu_Y,\\ &a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y\big)\end{aligned}}$$

For jointly Gaussian variables,

$$\boxed{\mathrm{Cov}(X, Y) = 0 \quad \iff \quad X, Y \text{ independent}}$$